

	External Assessment Specifications Document			
Curriculum Code	Qualification Title		NQF Level	
252901001	Occupational Certificate: Cybersecurity Analyst		5	
	Name	Email	Phone	Logo
Assessment Quality Partner	MICT SETA	Gugu.Sema@mict.org.za	011-2072600	

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DQP Representative Signature

Date

AQP Representative Signature

Date

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1 ASSESSMENT STRATEGY

1.1 Assessment Model

An external integrated summative assessment, conducted through the relevant QCTO Assessment Quality partner is required for the issuing of this qualification. The external integrated summative assessment will focus on the exit level outcomes and associated assessment criteria.

The assessment will be conducted through written assessment and the evaluation of practical tasks at decentralised approved assessment sites by a panel of assessors evaluated by assessor(s) registered with the AQP within a period of 1 day.

1.2 Qualification Purpose

The purpose of this qualification is to prepare a learner to operate as a Cybersecurity Analyst

Cybersecurity Analysts apply the practice of protecting assets such as networks, computer systems and information assets from malicious attacks and threats. They assess and mitigate risks and potential intrusions and identify risks and vulnerabilities. They study existing techniques for managing security issues and maintaining the security of information and systems in the working environment ensuring legal compliance

A qualified learner will be able to:

- Demonstrate knowledge and understanding of cybersecurity concepts and investigate how cybersecurity affects legal compliance and solidarity in companies and communities.
- Assess risk to assets and evaluate current cybersecurity protection measures.
- Implement detection, protection and prevention systems and respond to breaches or incidences.

1.3 Assessment Standards

1.3.1 Assessment Standards for the Qualification

Integrated Assessment Focus Area 1

Analyse, identify and solve potential and actual security risks, vulnerabilities and inefficiencies to safeguard information system assets from malicious cybersecurity attacks (30%)

Associated Assessment Criteria

- Regular checks are performed to ensure security practices are compliant with company policies and security requirements as well as the legal and regulatory requirements
- Critical systems and critical Information System Assets are identified
- Vulnerability testing is conducted to identify weaknesses, vulnerabilities and risks in hardware, software and IT infrastructures
- Data on current security measures are collected and leveraged for risk analysis and regular system status reports are written
- The optimum method of securing the IT infrastructure and Information Systems Assets of an organisation is identified
- Risk Catalogue is maintained and checked against incidence reports on a scheduled basis

Integrated Assessment Focus Area 2

Protect organization's digital assets from both internal and external threats by maintaining cybersecurity attack mitigation and incident response capability (30%)

Associated Assessment Criteria

- Anti-virus systems, firewalls, data centres and software defensive protocol updates are analysed and implemented with a security-first mindset
- Information system assets are safeguarded by identifying and solving potential and actual security problems
- Credentials are granted to authorized users, access-related activities are monitored and checks are done for unregistered information changes
- Security improvements are implemented by analysing current situation, evaluating trends and performing risk and threat analysis
- Intruder Detection and Prevention Systems and associated response protocols are implemented and automated

Integrated Assessment Focus Area 3

Execute ethical cybersecurity monitoring in line with cybersecurity's policies to provide defence to a level of confidentiality, integrity and availability equal with the threat to assets and their value to the company (15%)

Associated Assessment Criteria

- Security monitoring and incident response toolsets are applied to detect and trace intrusions and test the environment with a focus on continuous improvement
- Incidences are detected by identifying and analysing abnormalities and violations are reported
- Security vulnerabilities and inefficiencies are detected by conducting periodic checks and tests, and cybersecurity team is collaborated with to update defensive protocols as necessary
- A comprehensive understanding of ethics and a moral compass as applicable to cybersecurity is demonstrated

Integrated Assessment Focus Area 4

Execute response procedures to mitigate cyber-attacks and secure information assets, intellectual property and computer systems (15%)

Associated Assessment Criteria

- Unusual activities and attacks that were identified are monitored and responded to, appropriate defensive protocols are implemented if breaches occur and incidents are reported
- Response and mitigation activities are performed to close off the security vulnerability, prevent expansion of an event and to resolve the incident
- Data are collected on current security measures for risk analysis and regular system status reports are written
- Investigations into network intrusions and other cyber security breaches are executed

Integrated Assessment Focus Area 5

Execute recovery protocols and procedures as per recovery plan to restore data and/or assets affected by cybersecurity incidents (10%)

Associated Assessment Criteria

- The means of security breach is immediately found and the corrupted servers, devices and systems are quarantined to prevent spreading and to protect against additional vulnerabilities
- Information that was lost/stolen is identified, its impact is analysed and remediation procedures are started
- Attacks are analysed and cybersecurity improvements are developed based on lessons learned from such attacks as well as reviews of existing measures
- An incident close-out report is produced with appropriate logs and other substantiating evidence

1.3.2 Assessment Standards for Phases

None

1.3.3 Assessment Standards for Part Qualifications

None

2 ASSIGNMENTS TO BE EVALUATED EXTERNALLY

None

3 CRITICAL ASPECTS OF THE INTERNAL ASSESSMENTS TO BE ASSESSED EXTERNALLY

- None

4 CRITERIA FOR THE REGISTRATION OF ASSESSORS

Qualifications:

- Assessors and moderators: Accredited by the MICT SETA
- Subject matter experts: NQF 6 ICT related qualification

Experience:

- 1 year's experience in the IT industry
- 1 year's experience in assessment and moderation

5 FOUNDATIONAL LEARNING

Not required.

6 ELIGIBILITY REQUIREMENTS FOR THE EXTERNAL ASSESSMENT

In order to qualify for the external summative assessment learners must provide proof of completion of all required modules by means of a statement of results and statement of work experience as well as proof of successful completion of the following subjects and modules or alternative programs where applicable, including Foundational Learning Competence.

6.1 Qualification

	<i>Proof of Knowledge Modules</i>			<i>OR</i>			<i>Proof of Alternative Programmes</i>	
<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>		<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>
252901001-KM-01	Introduction to Cybersecurity	4	8					
252901001-KM-02	Fundamentals of Network Security and Defence	5	12					
252901001-KM-03	Cybersecurity and Cyber Threats and Attacks	5	12					
252901001-KM-04	Introduction to Cybersecurity Governance, Legislation and Ethics	4	4					
252901001-KM-05	Fundamentals of Design Thinking and Innovation	4	1					
252901001-KM-06	Refresher: Logical Thinking and Basic Calculations	4	3					
252901001-KM-07	Computers, Devices and Computing	4	6					

	Systems							
252901001-KM-08	Data and Database Vulnerabilities	4	3					
252901001-KM-09	Introduction to 4IR and Future Skills	4	4					

	<i>Proof of Practical Skill Modules</i>			<i>OR</i>			<i>Proof of Alternative Programmes</i>	
<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>		<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>
252901001-PM-01	Ensure Compliance in terms of Legal Cybersecurity Requirements and National and International Standards	5	4					
252901001-PM-02	Assess Risks and Vulnerabilities and Current Security Measures	5	20					
252901001-PM-03	Implement Protection, Prevention and Detection Measures to Mitigate Risk, Violations and Vulnerabilities	5	20					
252901001-PM-04	Apply Logical Thinking and Maths: Refresher	4	6					

252901001-PM-05	Apply Basic Scriptwriting for Cybersecurity Toolsets	4	4					
252901001-PM-06	Access and Visualise Structured Data Using Spreadsheets	4	5					
252901001-PM-07	Apply Design Thinking Methodologies	4	4					
252901001-PM-08	Function Ethically and Effectively as a Member of a Multidisciplinary Team	4	5					

And

Statement of Work experience

6.2 Phase 1

No Phase Tests applicable

7 PART QUALIFICATIONS

None